AI for Good Sandbox Hackathon – [Team Name] – MinorShield Policy Lab

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Github link: https://github.com/oumarouabdoulhamid26-svg/MinorShield-Hackathon

1. Introduction

MinorShield is a mobile parental-control app that uses artificial intelligence, directly on the child's phone, to detect dangerous content — nudity, violence, toxic text — and alert parents immediately, without ever transmitting the image or the child's identity. It is an AI-for-good solution for protecting minors online, and it is this project's use case. But deploying such a tool in Cameroon raises a question that has never been tested: when it detects a real crisis, such as a sexual-extortion ("sextortion") ring targeting school students, can the country's institutions lawfully receive its alert, coordinate, and act?

Such a crisis falls to no single institution: it mobilises a regulator, the telecom operators, a foreign platform, the police, the prosecutor, social services, schools, and the companies that hold children's data. Cameroon has adopted a Child Online Protection Charter (Law No. 2023/009 of 25 July 2023), but how its institutions would coordinate in applying it during a real crisis — triggered by a tool such as MinorShield — has never been rehearsed. Simulation exercises are rare, costly, and rely on how well participants remember the law.

To stress-test the legal environment around our use case, we built MinorShield Policy Lab, a multi-agent policy sandbox. Thirteen AI agents, each playing a real Cameroonian institution or private actor, handle a sextortion crisis triggered by a MinorShield alert and revealed in stages. Each agent knows only what has been disclosed to it, acts within the mandate the law gives it, and must justify every action with an article it actually retrieved from a national knowledge base. A legal-advisor agent judges the legal basis of each action, flags the gaps in Cameroonian law, and compares them with the European Union, the United Kingdom, the United States, and Australia. Every citation is verified automatically and every step is written to an audit log. The project thus answers directly to dimension D10 (AI and policy) of the AI readiness framework — using AI to experiment with policy before a crisis — and also touches D7 (strategic alignment), D4 (contextualisation and regional impact), and D8 (collaboration with AI).

2. Description of the use case

The use case. MinorShield continuously analyses, on the child's device, the screenshots and on-screen text: a vision model detects nudity and violence, a language model spots toxic speech, and the parent receives an alert with a score, without the image ever leaving the phone. The tool shares only aggregated, anonymised trends with the outside world. Such a system truly protects children only if, when it reveals a real crisis, the legal framework lets institutions receive the alert and act. It is this policy environment — not the app's code — that our sandbox puts to the test.

Gaps in existing approaches for stress-testing this environment. (1) Crisis exercises are rare, expensive, and leave no trace of why a decision was taken. (2) General-purpose chatbots answer legal questions from memory and invent plausible-sounding articles. (3) Comparisons with foreign law are done by hand, one country at a time. (4) Nobody can see in advance where mandates overlap or where no institution is competent.

Our solution answers each gap:

- **A citable knowledge base.** 11 official Cameroonian texts (Laws 2023/009, 2010/012 on cybersecurity, 2010/013 and 2015/006 on electronic communications, 2010/021 on e-commerce, and six implementing decrees including those organising ANTIC, ART and MINPOSTEL), taken from the 2023 official compendium published by the telecom regulator; 5 foreign references (EU Digital Services Act, GDPR, UK Online Safety Act 2023, US COPPA Rule 16 CFR 312, Australian Online Safety Act 2021); and the 64 strategy and use-case documents supplied by ITU. Legal texts are split **article by article** with their page

number, so each extract is a citable unit (7,740 indexed chunks in total). Provenance of every file is documented.

- **Role agents with real mandates.** Each of the 15 agents (13 actors, 1 legal advisor, 1 neutral facilitator) has a role card: mission, legal basis, limits, priority sources. Mandates were written from the articles of Law 2023/009 (e.g. art. 27–29 for the ICT regulator ANTIC, art. 32 for police and prosecutors, art. 33 for internet access providers).
- **Unequal, sequenced information.** The scenario "Operation Mirror" has 5 injects over 3 days, each revealed only to the concerned institutions: a 400% rise in blocked content detected by MinorShield; two victims (aged 15 and 19), one facing school exclusion; images spreading on a foreign platform ("Tchatlink", fictitious); a data leak from a school-management app ("ScolaPlus", fictitious) used by 300 schools; and a proposal to block the platform nationwide.
- **Coordination by requests.** Agents address requests to each other; the facilitator routes them and returns the answers to the sender's inbox, so information spreads only through exchanges.
- **Verified citations.** Each legal claim receives a status: supported (cites an extract actually provided), doubtful (mentions an article, a named law or a jurisdiction absent from the cited extract), unsupported (no citation) or invalid (cites an extract that was never provided, i.e. an invention).

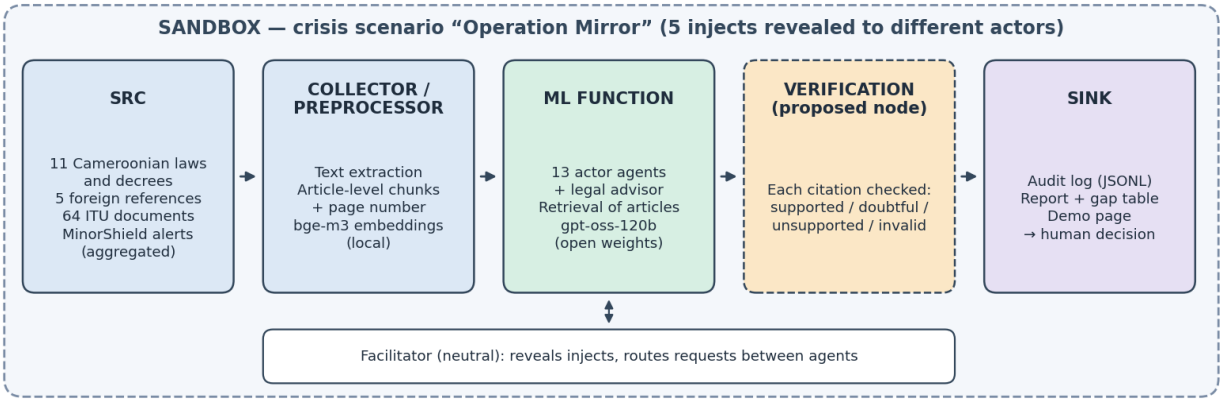


Figure 1 – Architecture of MinorShield Policy Lab mapped onto the ITU-T Y.3172 pipeline

3. Use case requirements

- **Source data (SRC):** legal texts must come from official publications, with the exact source, date and page recorded; MinorShield may only share **aggregated, anonymised** trend data, never a child's content or identity.
- **Collection and preprocessing:** native text extraction (OCR only as a cross-check of a scanned copy); headers and footers removed; article-level chunking that rejects false article references; country and document-type tagging; multilingual embeddings computed **locally** (bge-m3 via Ollama), so no legal corpus leaves the machine.
- **Model:** an open-weight reasoning model (gpt-oss-120b, served by Groq), structured JSON output (evaluation, actions, legal basis, requests, gaps, confidence, explicit reasoning). Each request must fit the free-tier limit of 8,000 tokens per minute, so the context is fitted automatically (at most 4 extracts of 600 characters per agent; the advisor keeps one extract per foreign jurisdiction).
- **Policy rules:** an agent acts only on information revealed to it; it may cite only the extract IDs it received, never an article from memory; it must state what the law does not allow it to do.
- **Sink and human oversight:** outputs (audit log, report, demonstration page) are deliberation material for human decision-makers, not legal advice. Every run records the model, tokens and duration for reproducibility; the reference run used 159,930 tokens in 19 minutes on a CPU-only laptop.

4. Mapping with ITU-T Y.3172, the AI Readiness framework and domain-specific policies

The table maps each node of the ITU-T Y.3172 machine-learning pipeline to our implementation, to the dimensions of the ITU AI Readiness framework, and to the documents in our knowledge base that set the corresponding requirements.

Node	Use case (child online protection crisis)	Readiness 2.0 Report	Documents (policies, strategies, regulations)
SRC	Official Cameroonian laws and decrees; foreign references; ITU strategy corpus; MinorShield aggregated alerts	D1 Data/Model Marketplace (dataset properties, data sources, provenance); D4 Contextualization	Law 2023/009 (Child Online Protection Charter); Law 2010/012 (cybersecurity); Decree 2019/150 (ANTIC); Law 2024/017 (data protection, text not yet available)
C	Ingestion scripts: PDF/HTML to text, splitting of the official compendium, incremental addition	D3 Cross-Domain Correlation (law, strategies, use cases); D13 Digital Infrastructure (CPU-only laptop)	ITU reference code (ITUAIReadiness, MIT licence); provenance record (bd/SOURCES.md)
PP	Article-level chunking with page; country and type tagging; local bge-m3 embeddings; vector store	D1 metadata; Standards factor (interoperable, citable units)	ITU-T Y.3172 (preprocessing); official article numbering of each text
M	15 agents on open-weight gpt-oss-120b, retrieval-augmented, structured JSON with explicit reasoning	D5 Integration in Workflows; D9 Impact of Humans (role cards written from legal mandates); Open Source factor	G7 Hiroshima Code of Conduct; Singapore Model AI Governance Framework for GenAI 2024; EU AI Act (summary)
P	Information visibility per agent; cite-or-flag rule; mandate limits; advisor verdicts; automatic citation check	D10 AI & Policies; D8 Collaboration with AI	Law 2023/009 art. 4–10, 14, 27–33 (mandates and sanctions); UN System White Paper on AI Governance 2024
D	Facilitator routes requests between agents (max. 4 per inject) and returns the answers	D7 Strategy Alignment; D6 Human Interface	Y.3172 interfaces; no formal Cameroonian crisis coordination body (identified gap)
SINK	Audit log (JSONL), readable report, self-contained demonstration page, gap table vs EU/UK/US/AU	D10 AI & Policies; D11 AI for Inclusion; D12 Granular Priorities (age groups 15 vs 19)	EU DSA; GDPR (art. 33); UK Online Safety Act 2023; US COPPA Rule; Australian Online Safety Act 2021
Sandbox	The crisis simulation itself: policies are tested with no real-world consequence	Sandbox factor; D10 (policy experimentation)	ITU AI for Good Sandbox; AI Ready Framework 2025

Beyond data and model handling, the mapping covers **strategy alignment** (D7: 13 institutions acting on one shared legal base), **human interface** (D6: a single-file demonstration page where each decision opens the exact article it relies on), **collaboration between AI and humans** (D8: agents state their confidence and their gaps; humans decide), and **inclusion** (D11: French-language legal corpus, free open-weight models, runs without a GPU).

5. Error scenarios

Reference run (5 injects, 56 agent interventions, 20 routed requests). 124 legal claims were checked automatically: **109 supported (87.9%)**, 12 without citation (mostly practical actions such as preserving evidence; two agents explicitly stated that no extract authorised the requested action), 2 doubtful and 1 invalid. The advisor judged 22 actions (10 founded, 6 to be verified, 6 unfounded) and identified 10 gaps, each compared with at least one foreign jurisdiction. The full log and report are published with the code.

How the injects were handled.

- **Inject 1, early signal.** MinorShield reported aggregated trends to ANTIC. The advisor found no legal procedure or alert threshold for a private child-safety tool to share data with the authorities.
- **Inject 2, victims and school exclusion.** MINAS and MINPROFF asked the school not to exclude the 15-year-old victim; the police qualified the offence under the cybersecurity law. **Missed:** all agents treated the

19-year-old as a minor, although Law 2023/009 (art. 3) defines a child as a person under 18; she falls outside the Charter while still a legal minor under civil law (majority at 21).

- **Inject 3, foreign platform.** Agents relied on art. 14 (suspension of a user at the request of the authorities), and ANTIC planned a formal notice under art. 27–28. The advisor noted the absence of removal deadlines for foreign platforms, but **no agent noticed** that the art. 28 notice period, 15 days, is far too long for viral content.
- **Inject 4, data leak.** Agents concluded that no breach-notification procedure exists. In reality the data protection law (No. 2024/017) exists but its full text is not available online and therefore not in the knowledge base: a **gap in the knowledge base was read as a gap in the law**.
- **Inject 5, controversy: blocking the platform nationwide.** The advisor found no legal basis for blocking a whole platform and recommended creating that power. Only the platform agent raised proportionality. Our answer to the controversy: the system never decides; it produces the material for a human deliberation, which must weigh freedom of expression and economic impact, and favour targeted measures (art. 14 user suspension, removal of specific content) under judicial control before any general block.

AI errors caught, and not caught. The verifier flagged an invented extract ID and a US state law (CCPA) cited although it is not in the base. It cannot detect a correct citation used to support a wrong claim: the advisor attributed "financial transparency" duties to COPPA and overstated the Australian blocking powers. Against the 10 gaps we wrote into the scenario beforehand, the run found 5 fully, 3 partially and missed 2. These measures make the limits visible, which is why a human legal review remains mandatory.

6. Policy gaps and standards gaps derived from the mapping exercise

Policy gaps (Cameroon) compared with global practice:

- **Reporting by private actors:** Law 2023/009 obliges internet access providers to report content harmful to children to the authorities (art. 33, fines of 25 to 250 million FCFA), but sets no channel or alert threshold for a third-party child-safety tool or for aggregated, anonymised signals. The UK Online Safety Act 2023 requires providers to report child sexual exploitation and abuse content to the National Crime Agency.
- **Removal speed:** a 15-day formal notice (Law 2023/009, art. 28), against 24 hours under the removal notices of the Australian Online Safety Act 2021.
- **Foreign platforms:** limited extraterritorial reach. The EU Digital Services Act applies to services offered to EU users wherever the provider is established and requires a legal representative in the EU.
- **Data breaches involving children:** no accessible notification procedure, against 72 hours under GDPR art. 33; the data protection authority has existed only since June 2026.
- **Age threshold:** victims aged 18 to 20 are outside the Charter although they are civil minors.
- **Mandates and coordination:** overlapping missions of MINAS and MINPROFF, no school directive for victims, and no formal inter-agency crisis coordination body; the facilitator had to be neutral for this reason.
- **Platform blocking:** no legal framework, hence no proportionality safeguards, for measures beyond targeted suspension.

Standards gaps:

- **Machine-readable law.** Cameroonian texts exist only as PDF compilations, without a consolidated, article-level open format (such as Akoma Ntoso); every legal knowledge base must be rebuilt by hand.
- **Auditing AI legal reasoning.** Y.3172 has no node for checking that an output is grounded in its sources. We propose adding a verification step between the ML function and the sink, using the four citation statuses above and two metrics: the citation support rate and the recall of expected gaps.
- **Content fidelity.** No automatic check yet verifies that a cited article really says what the agent claims; a standard human-review protocol for AI policy sandboxes is needed.
- **Reference corpus quality.** Two PDFs in the ITU reference repository are truncated (Kenya strategy, China framework), and 20 of 67 documents were tagged with the wrong country; both were corrected and documented in our repository.